

Sourdough at Sea — The Wild-Yeast Loaf

A living culture, a long cold proof, a hot covered bake. The wild loaf made reliable on a moving galley — and the acidity that flavours it also makes it keep.

Yield: **2 loaves** (10–12) Active: **40 min** Total: **2–3 days** Oven: **250–230°C** (covered) Done: **96–99°C** internal

WHERE IT WAS BORN The oldest leaven there is — wild yeast in flour and water, the <i>only</i> way to raise bread before industrial yeast in the 1860s. Egyptian loaves >3,000 yrs ago; the name from Klondike prospectors who guarded the crock through winter.	EATEN, AUTHENTICALLY The everyday country loaf of every grain culture — torn warm with oil or butter; San Francisco gave it a famous tang and a bacterium named for the city.	HOW IT’S USED NOW The craft-bakery boule — and the chef’s living culture, carried boat to boat down the dock. Crumb for brunch, toast, bruschetta; stale, it makes <i>panzanella</i> and ribollita.
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INGREDIENTS

2 LOAVES · WEIGHT

LEVAIN (NIGHT BEFORE)

Mature starter (<i>ripe</i>)	0.71 oz
Bread flour	$\frac{3}{4}$ cup
Water (<i>bottled / dechlor.</i>)	$\frac{3}{8}$ cup

FINAL DOUGH

Bread flour (12–13%)	6 $\frac{3}{8}$ cups
Wholemeal / rye	$\frac{3}{4}$ cup
Water, cool (<i>autolyse</i>)	2 $\frac{1}{2}$ cups
Water (<i>reserved, w/ salt</i>)	3 $\frac{1}{3}$ tbsp
Fine sea salt	1 $\frac{1}{8}$ tbsp
Levain (<i>all, at peak</i>)	~7.1 oz

SHAPING & BAKING

Rice flour (<i>banneton</i>)	3 tbsp
Semolina / flour (<i>dust</i>)	—

THE RATIO — FLOUR = 100%
Water 75% · Levain 20% · Salt 2%. Four numbers; everything above is one batch of them, scaled for two crew loaves. Learn them and you never need the recipe again.

Kit: Dutch oven or combi/Rational · 2 bannetons (or towel-lined bowls) · lame / razor · scale · bench scraper · mixing tub · probe thermometer

METHOD

- Build the levain.** Night before: 20g starter + 100g flour + 100g water. 4–8 h until doubled and a spoonful **floats**.

A purpose-built levain gives a known, vigorous population at a known time; the float test reads enough gas to lift the loaf. — Robertson; Modernist Bread
- Autolyse.** Mix 800g bread + 100g wholemeal flour with 600g water to a shaggy mass — **no levain, no salt**. Rest 45 min.

Flour hydrates and gluten knits with no work; salt and acid now would tighten it too early. — McGee; López-Alt
- Add levain, then salt.** Pinch in the ~200g levain; dissolve 20g salt in the reserved 50g water and work through.

Dissolved salt spreads evenly and reins the yeast to a pace the long proof can ride. — Forkish
- Bulk + 4 folds.** 24–26°C, ~4–5 h. Four stretch-and-folds in the first 2 h. Stop at ~40% risen, domed, jiggly.

Folds build aligned gluten that traps big bubbles — the open crumb. Judge by rise, not the clock; a warm galley halves it. — Robertson
- Pre-shape & shape.** Cut in two, pre-shape, bench 25 min, shape taut boules, **seam-up** in rice-floured bannetons.

The rest stops tearing; a tight skin holds the loaf through proof and bake. Rice flour has no gluten, so it won’t glue. — Forkish
- Cold proof 12–36 h.** Bag and into the fridge at 4°C. Bake straight from cold, no warm-up.

Yeast crawls while bacteria build acid — deeper flavour, blistered crust, clean scoring; and it uncouples bake from proof so you fire to order. — Modernist Bread

Bake: preheat Dutch oven/combi **250°C**. Turn out cold, **score** one slash, **lid on 20 min**; lid off, **230°C**, 20–25 min to deep mahogany / **96–99°C** internal. Cool **≥2 h** before cutting — the crumb sets as it cools.

Keep it alive at sea — the catch is the water. Chlorine, chloramine and stripped water-maker product stall a culture — feed bottled spring water, or stand tap water uncovered overnight. Warm galley (26–30°C) ferments fast (feed more, use cooler water); a cold fridge holds it dormant a week+ between feeds.

CARRY IT ANYWHERE — DEHYDRATE

Smear ripe starter **paper-thin** on parchment, dry **below 40°C** (dehydrator, low oven, shaded deck), flake and **vacuum-seal**. Survives months and flights in a pocket; revive over two or three feeds. *Insurance against a crossing crew who bins your jar.*

ELEVATION

TIER 1 — NO EXTRA TIME

- Stiff levain (60% water) — milder, sweeter
- Longer cold proof — more tang, to 36 h
- Tin + foil lid — when shaping or sea won't cooperate

TIER 2 — WORTH IT

- Country loaf — 20–30% wholemeal/rye, +water
- Seeded — 150g soaked seeds after the salt
- 80–85% hydration — wilder, holier crumb

TIER 3 — SERVICE

- Olive & rosemary laminated in the folds
- Bâtard — more crust, easy sandwiches
- Discard → crackers, crumpets, blinis

TROUBLESHOOT	
Dense, tight	Levain not ripe / under-proofed → float test; bulk to ~40% domed
Flat, spreads	Over-proofed / weak shape → catch bulk earlier, shape taut, proof <36 h
Gummy centre	Pulled pale → bake to 96–99°C; cool 2 h before cutting
Too sour	Levain old / fridge warm → use younger, stiffer levain; verify 4°C
Pale, no spring	Pot/oven cool → true 250°C, deeper score, lid on 20 min
Starter sluggish	Chlorine / water-maker water → bottled water, warm corner

CHARTER PREP & STORAGE	
Bake to order	Hold shaped loaves 36 h at 4°C; fire from cold as the table needs
Par-bake	Covered to pale (~25 min); finish 10–12 min at 230°C at service
Frozen baked	1 mo wrapped; refresh 10 min at 180°C from frozen — freezes better than yeasted
Room temp	Cut-face down, in cloth, 3–4 days — never the fridge (stales fastest)
Carry the culture	Dehydrate & vacuum-seal — months, flights, pocket-sized